

Project Proposal: Taskly

Overview:

Taskly is an interactive task manager built in *Processing* that transforms productivity into play. Instead of plain checkboxes, users complete daily tasks to earn points, feed a virtual pet, and unlock new evolutions. The program blends organization with simple game mechanics, making productivity feel rewarding rather than repetitive. It's designed for kids—or anyone who wants to make their schedule feel more alive.

Core Features:

- **Custom Task Creation:** Users can type new tasks, assign time limits, and mark them complete.
- **Interactive Interface:** Each task appears in its own animated box, visually aligned to form a personalized schedule.
- **Gamified Progress:** Completing tasks grants points that can be used to feed a virtual pet, causing it to grow or evolve.
- **Pet Evolution System:** The pet evolves through several stages based on the user's productivity.
- **Spin Wheel Mini-Game:** After completing tasks, players can spin a colorful wheel to earn bonus food or points.

Goal:

To design an engaging way to encourage organization, discipline, and fun using creative visual feedback and interactive programming concepts.

Roadmap

Phase 1: Setup and Core Mechanics

1. Set up your window and basic layout (`size()`, background color, etc.).
2. Create a basic `Task` class with properties like `title`, `done`, and `timeLimit`.

3. Display a few test tasks on screen using rectangles and text.
4. Implement `mousePressed()` to toggle tasks on/off.

Phase 2: Add Game Elements

5. Add a simple `Pet` class with a few “growth stages.”
6. Display the pet and have it grow when tasks are completed (based on points).
7. Add points logic: each task completed gives a certain amount of points.

Phase 3: Interactivity & Animation

8. Animate task boxes (like a bounce when checked).
9. Add the spin wheel feature using randomization and rotation math.
10. Create a “reward” system tied to the spin results (extra food, points, etc.).

Phase 4: UI Polish and User Input

11. Implement a simple text input or button-based task adder.
12. Improve visuals with `PImage`, custom colors, and transitions.
13. Add sound effects or background music

Phase 5: Refinement & Extra Features

14. Add file saving (`saveStrings`, `loadStrings`)
15. Optimize code with functions and organized classes.
16. Polish animations, fix bugs, and finalize visual design.

Journal

11/4

Goals: Finalize project proposal, and what game I am going to make

My main goal today was to wrap up my project proposal and settle on the idea I wanted to build. After going back and forth, I committed to Taskly. It took a while to decide whether I wanted something fun or something practical. Choosing one meant giving up the other, which made the decision harder than I expected. Once I finally chose Taskly, I started laying down the first bits of code. Those early steps always feel clunky because nothing looks right yet, but it felt good to at least get the project moving.

I had problems deciding whether to do a fun or practical project for my coding game. Weighing and choosing both sides was particularly hard, and doing the first few steps to get the code up and running was challenging as well.

Notes: Start going into the coding process -- finalize more details for Taskly and a gist of how. It is going to work

11/6

Goals: Continuing programming, start working on classes and functions → get some of the way done with drafting stage

My goal for the day was to keep pushing forward with the programming side, especially the classes and functions. I finished a mousePressed function and added a few booleans. The beginning was rough because I couldn't get mousePressed to talk to my boolean system the way I wanted. Once I figured out how they connected, everything clicked a little better and I could build from there.

Notes: Continue with coding -- start working on the spinny wheel and different unique designs for pets and how they should grow

11/10

Goals: Start working on classes and functions - continue improving last times' problems.

- Use AI to fix errors, structure, and easier code functions.

Today I managed to create another variable of the pet size, using moused pressed functions I can now increase the pet size once you are able to tap on it.

I had some huge problems with my code and had many errors. I had to spend the first half of class fixing my code. However, I stayed persistent and asked for help along the way and took initiative to find my own errors. I fixed it and continued working on my functions and moused

pressed and I am hoping to finish my draft in the next few classes before starting the actual project.

Notes: Finish working on the outline, and get deeper into the coding processes.

11/12

Goals: Finish solving the errors from last class and continue working on the classes and functions to finish my draft. I am hoping to finish my first rough draft by the middle of next week.

I ended up getting more done than expected. I added a point system to make Taskly more fun to use, and I finally got the spin wheel to generate random points that feed into the total. At the start of class, I wasn't sure how to pick up from where I left off, but once I found my footing, everything came together, and I got a lot done.

I faced some challenges at the beginning of class on how to continue with where I left off, but after that, I was extremely focused and got a lot of things done.

Notes: Continue working on the class system, make code and Taskly more engaging.

11/14

Spent today's entire class with Mr.Cloos trying to debug the spin wheel glitch

Notes: I spent the whole class debugging the spin wheel glitch with Mr. Cloos. It was a long process, but it helped me understand what was breaking the rotation.

Next class continue to work with Mr.Cloos on the Push and Pop matrix

11/18

Notes: I finally finished fixing the code, and everything is working smoothly again. Now I can focus on adding new features and keeping up with the rest of my plan.

- Focus on finalizing the most important parts

11/20

Going to finish working on the spin wheel

→ Used AI to understand what translating, push and pop matrices mean.

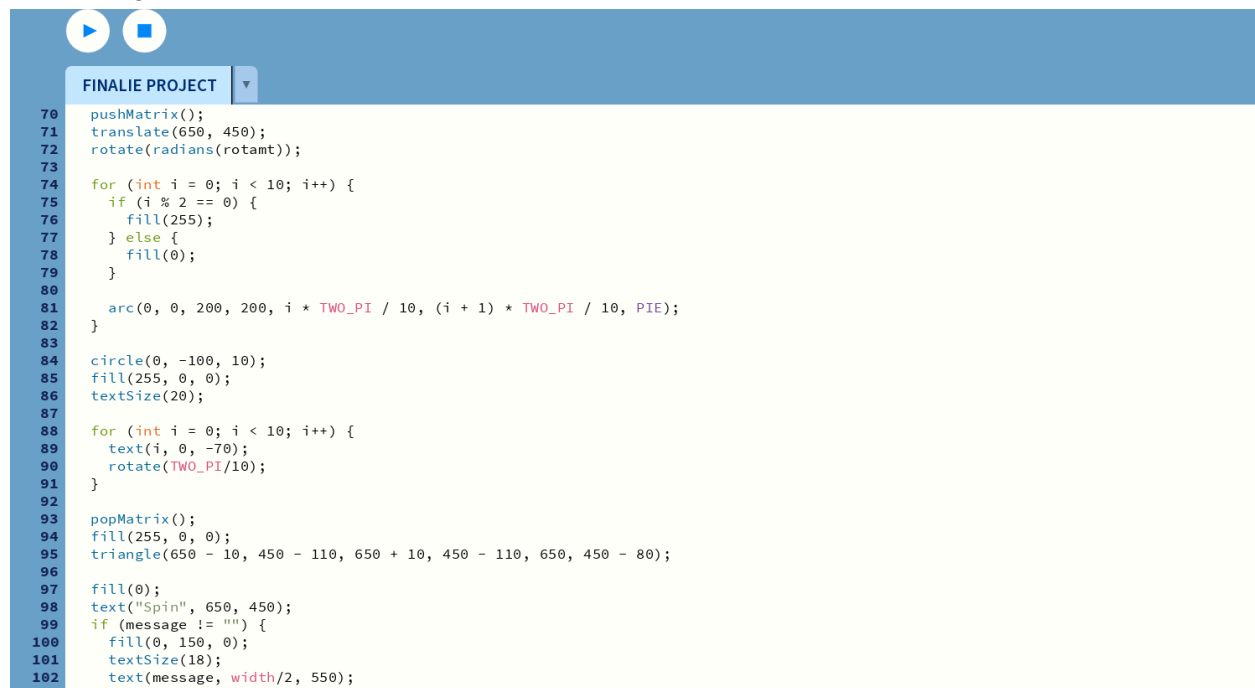
Origin point is 0,0 on the corner of the screen which results in the circle moving a quarter off the screen. Translate allows the point of origin to be seen by moving into the middle to allow the full circle to spin. The Push and Pop matrix allows the words in the previous screen to be translated down as well.

It was difficult in the beginning to understand the concept of the Push and Pop Matrix as it is one of the most monotonous and difficult concepts to understand in processing (Java). However, I used AI and coding bat to practice and master the Push and Pop Matrix.

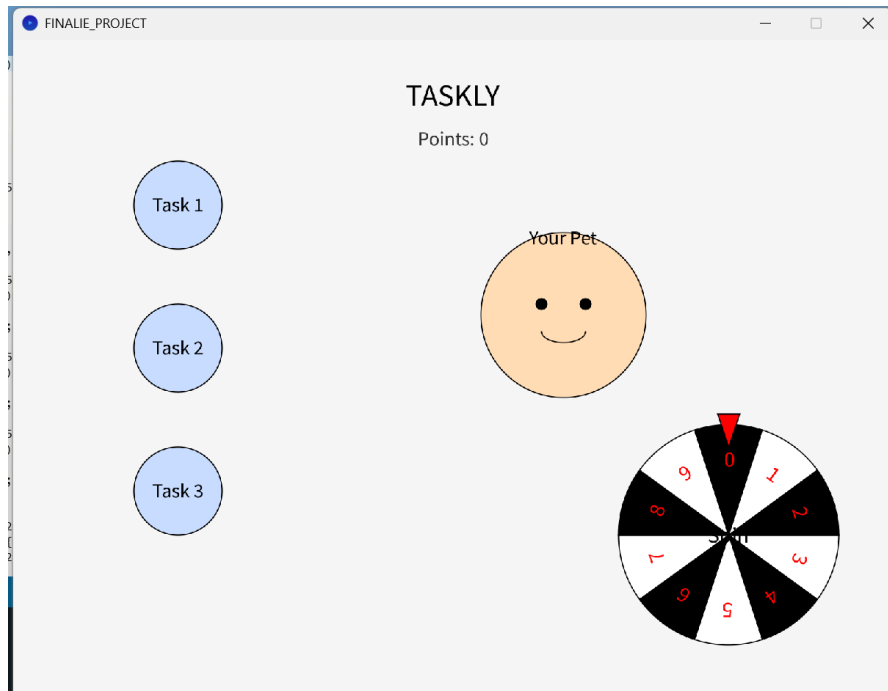
Continue to work on the code and add new creative things to the pets and tailor the tasks more to the user as well as finding a way to keep track of the score forever with a restart button.

11/24

Continued to work on my spin wheel and found out many different methods on how to rotate in processing

A screenshot of a Processing IDE window. The title bar at the top says "FINALIE PROJECT" with a dropdown arrow. Below the title bar are two circular icons: a play button and a square. The main area is a code editor with a light yellow background and blue line numbers on the left. The code is in Java and uses Processing's P5.js library. It defines a spin wheel with 10 segments, alternating between white and light blue fill. The wheel is centered at (650, 450) with a radius of 200. The code includes functions for pushing and popping the matrix, translating to the wheel's position, rotating by a specified angle, and drawing the segments. It also includes a text label "Spin" at the center and a message box that appears when the wheel is clicked.

```
70 pushMatrix();
71 translate(650, 450);
72 rotate(radians(rotamt));
73
74 for (int i = 0; i < 10; i++) {
75   if (i % 2 == 0) {
76     fill(255);
77   } else {
78     fill(0);
79   }
80
81   arc(0, 0, 200, 200, i * TWO_PI / 10, (i + 1) * TWO_PI / 10, PIE);
82 }
83
84 circle(0, -100, 10);
85 fill(255, 0, 0);
86 textSize(20);
87
88 for (int i = 0; i < 10; i++) {
89   text(i, 0, -70);
90   rotate(TWO_PI/10);
91 }
92
93 popMatrix();
94 fill(255, 0, 0);
95 triangle(650 - 10, 450 - 110, 650 + 10, 450 - 110, 650, 450 - 80);
96
97 fill(0);
98 text("Spin", 650, 450);
99 if (message != "") {
100   fill(0, 150, 0);
101   textSize(18);
102   text(message, width/2, 550);
```



I kept working on the spin wheel and tried several different rotation methods in Processing. Each one worked a little differently, so testing them helped me understand how the transformations operate.

11/26

Today I was very productive on creating a new `mousePressed` function which allows me to grow the size of the pet whenever your tasks are completed.

Some problems I encountered were the distances and `!=` function was new to me so I had to learn that. However, after I got the hang of doing one, I copied and pasted and just put the numbers in.

I wanna start tailoring the pet to make more unique characteristics and keep on growing my code.

```

132
133 void mousePressed() {
134     if (dist(mouseX, mouseY, 150, 150) < 40) {
135         task1Done = !task1Done;
136         petSize += task1Done ? 10 : -10;
137         points += task1Done ? 10 : -10;
138     }
139
140     if (dist(mouseX, mouseY, 150, 280) < 40) {
141         task2Done = !task2Done;
142         petSize += task2Done ? 10 : -10;
143         points += task2Done ? 10 : -10;
144     }
145
146     if (dist(mouseX, mouseY, 150, 410) < 40) {
147         task3Done = !task3Done;
148         petSize += task3Done ? 10 : -10;
149         points += task3Done ? 10 : -10;
150     }
151     if (dist(mouseX, mouseY, 650, 450) < 80 && !spinning) {
152         spinning = true;
153         rotateby = 10;
154         message = "";
155     }
156 }
157

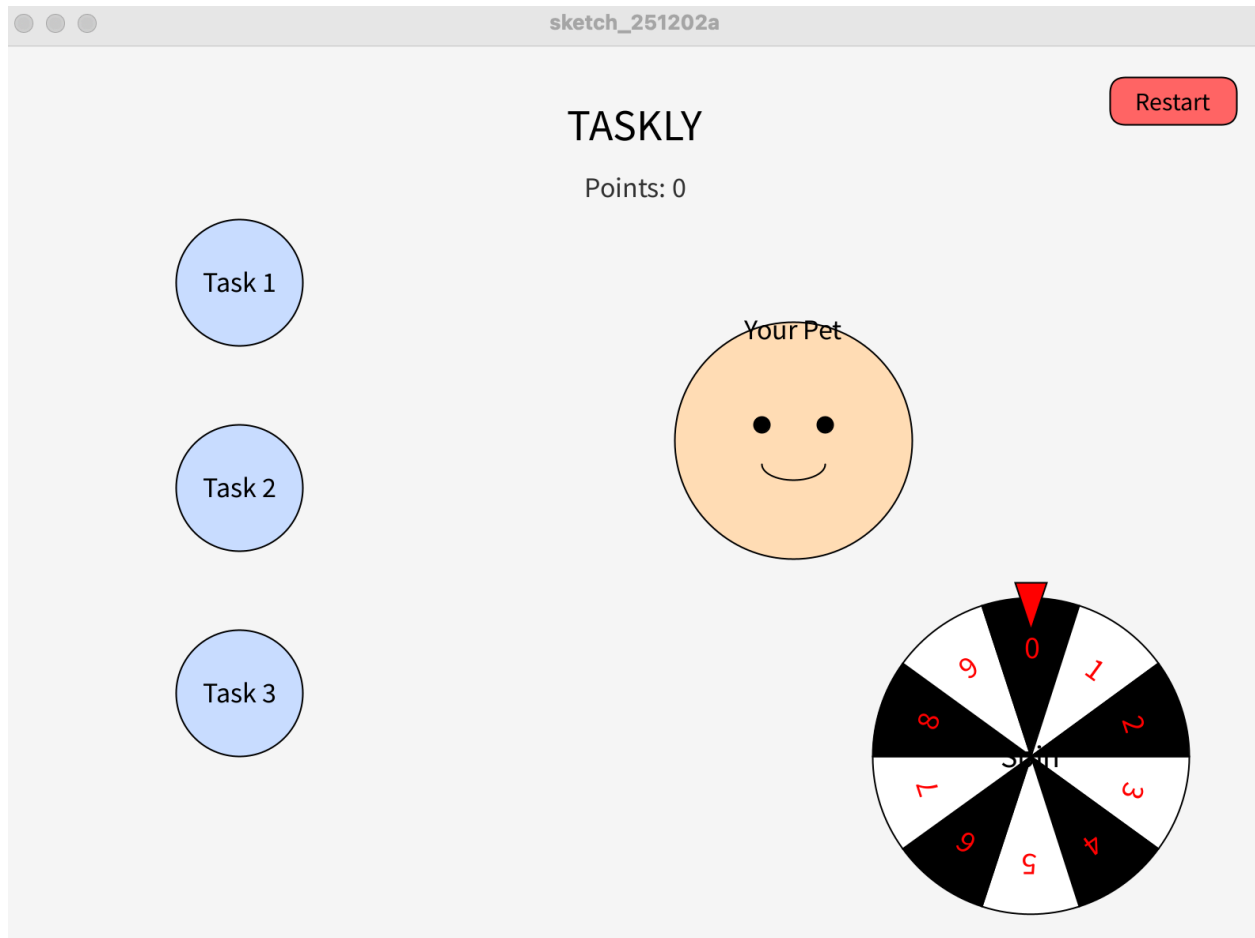
```

12/2

I was very productive in my work today as I added a crucial element to my game. Allowing my points to save progress, and make it possible to restart the points using new Void functions.

I had encountered some problems with my original code, and remembered what I was doing before Thanksgiving break. However, my journal came in handy and led me right where I left off.

Next class, I want to focus on completing my final touches to the game and maybe add extra details for tasks, and design for my pet.



12/4

I spent today polishing the code and making small adjustments as I prepare the final draft. I'm also reviewing the push and popMatrix concept again just to make sure I fully understand it. Notes: Make sure to get the code up and running as it is the final week to make adjustments

12/8

I looked at the spin wheel examples Mr. Cloos shared and experimented with new ways of making the wheel rotate. Taskly has come a long way. The game runs well, the major features are solid, and all the previous errors are fixed. It feels like the project finally turned into the idea I had in my head when I wrote the proposal.

12/10

Today was the last full day of the project, and it felt like everything came together. When I first chose Taskly, I wasn't completely sure whether the idea would actually work the way I imagined. Looking back, I'm glad I pushed through the early confusion, because the final version feels very genuine to me. I spent a little time testing everything again and fixing a few tiny details that I hadn't noticed before. Once the code was running the way I wanted, I looked through my journal and realized how much time I had spent. The spin wheel alone took multiple classes and several rounds of debugging, but now it's one of the most polished parts of the game. The pet

system, the point saving, the mousePressed functions, and even the push/pop matrix work all came out of moments where I got stuck and had to figure my way through it.

Reflection: Now that Taskly is finished, I can see how much more confident I've become at reading errors, structuring functions, and thinking through design choices. I'm proud of how the game turned out, and even more proud of how much I grew throughout the process. If there were more time, I'd keep expanding the pet designs and add more personalization for the user.